NATIONAL UNIVERSITY OF COMPUTER & EMERGING SCIENCES

SEMESTER PROJECT — PHASE 2 GUIDELINES

AI4015: Agentic Artificial Intelligence

Phase 2 — Development, Evaluation & Final Submission

Deadline	Sunday, 8 November 2026, 11:55 PM (PKT)
Submitted by	Group head only, one submission per group, through Google Classroom
Deliverables	(A) Final Source Code & Documentation | (B) Research Paper (PDF & Source)
Builds on	Your Phase 1 base paper, selected research gaps, proposal, and baseline
Supervisors	G1 projects: Muhammad Sajjad; G2 projects: Meemon Khan; G3 projects: Naeem Tareen

1. Purpose of Phase 2

Phase 2 is the main implementation and research phase. Each group turns its Phase 1 baseline into a complete agentic system, evaluates it scientifically against the base paper, and reports the results as a conference-style research paper. The central question of this phase is simple: for each of your selected research gaps, did your system improve on the base paper, by how much, and why? Every part of this phase, from the code to the paper, should help answer that question with empirical evidence.

2. Starting Point: Phase 1 Feedback

- Address all feedback given by your project supervisor on Phase 1 before building on it.
- The base paper and the selected research gaps are fixed after Phase 1. Any change requires written approval from your project supervisor.
- If your design changed during development (e.g., a different model, tool, or dataset split), record the change and the reason in a short "Changes since Phase 1" section of the repository README.

3. Deliverable A — Final Source Code

3.1 Complete Agentic System

Build the complete system on top of the Phase 1 baseline. The system must show genuine agentic operation, not a set of isolated components. It must include:

- **Complete pipeline:** An end-to-end agentic workflow executing on selected data, environment, or benchmark.
- **Gap-specific components:** For each selected gap, a clearly identifiable component or module that addresses it, switchable for ablation.
- **Environment interaction:** Meaningful interaction with tools, data, APIs, software, simulators, or environments.
- **Observable behaviour:** Logged agent trajectories demonstrating reasoning steps, tool calls, inputs/outputs, and state transitions.
- **Robustness:** Failure handling, verification, retries, recovery, or human-in-the-loop control where required.
- **Iterative improvement:** Documented progression showing how observed failures informed algorithmic and system refinements.

3.2 Development Log

Maintain a concise development log in the repository (e.g., `docs/devlog.md`). For each major iteration, record the date, the failure or limitation observed, the change made, and its measured effect.

3.3 Repository Requirements

The final GitHub repository must allow your supervisor to install the system, run it, and reproduce your primary findings:

- Clean, modular, and documented codebase with an informative README.
- Reproduction scripts and commands for experiments (E1–E6) saving structured results (e.g., in `results/`).
- Agent traces and trajectory logs for reported runs.
- A usable application interface (CLI, web UI, or API) or reproducible execution workflow.
- Unit and integration tests for deterministic modules.
- Dependency configurations (e.g., `requirements.txt`, `pyproject.toml`, or `Dockerfile`) and `.env.example`. Never commit secrets.
- Balanced commit history from all group members.

> **Release Tag:** Create a GitHub release or tag named `phase2-final` before the deadline. Only code within this release will be assessed.

Folder / File	Contents
`src/`	Agent, tools, state, orchestration, and gap-specific components
`data/`	Data loaders and split definitions (or download scripts for large datasets)
`experiments/`	Configurations and reproduction scripts for E1–E6
`results/`	Saved results, metrics, tables, and figures used in the research paper
`logs/`	Agent trajectories and run logs (or representative samples)
`tests/`	Automated test suites for deterministic components
`docs/`	Development log, architecture diagram, and extended documentation
`README.md`	Overview, environment setup, execution steps, and gap-resolution table

4. Experimental Evaluation

Evaluation must run alongside development rather than being deferred to the conclusion of the project. A system without thorough empirical validation remains incomplete.

4.1 Experimental Setup

- **Data splits:** Use fixed splits. The test set must never be used for prompt tuning, parameter optimization, or design decisions.
- **Reproducibility parameters:** Explicitly log model versions, decoding temperatures, system prompts, and random seeds.
- **Statistical rigor:** For stochastic processes, execute each main evaluation at least three times, reporting the mean and standard deviation.

- **Fair benchmarking:** Benchmark directly against the base paper under identical splits and evaluation metrics. Explicitly identify reproduced versus paper-reported figures.

4.2 Required Experiments

#	Experiment	Objective & Description
E1	Baseline	Performance of the simpler or non-agentic baseline defined in Phase 1.
E2	Base paper	Performance of the base paper's approach on identical data and metrics (reproduced where possible).
E3	Proposed system	Performance of your full, unablated agentic system across the evaluation suite.
E4	Gap-wise experiments	Targeted experiments evaluating hypotheses formulated for each identified research gap.
E5	Ablation study	Systematic removal or substitution of individual gap-specific modules to quantify respective contributions.
E6	Efficiency & reliability	Quantitative metrics covering financial cost, latency, token throughput, tool calls, and failure recovery rates.

4.3 Failure Analysis

Categorize observed system failures on the evaluation set into distinct failure modes. Report frequency distributions, present at least two representative success trajectories and two representative failure trajectories, and provide technical post-mortems for observed failures.

4.4 Gap-Resolution Table

Summarize findings using the format below. This table must appear in both the repository README and the research paper.

Gap	Method / Component	Metric	Base Paper Result	Our Result	Addressed? (Fully / Partially / Not)
Gap 1					
Gap 2					

> **Reporting Integrity:** Negative or partial results are fully acceptable and will not be penalized if rigorously documented, analyzed, and reproducible via the codebase.

5. Deliverable B — Conference-Style Research Paper

The final report must be written as a formal conference paper conforming to the standards of peer-reviewed technical venues: objective tone, precise technical vocabulary, robust logical progression, and full empirical support.

5.1 Format Specifications

- **Length & Style:** 8–10 pages (inclusive of references) formatted in two-column IEEE conference style (`IEEEtran`) or ACM conference style (`acmart`). Overleaf / LaTeX is recommended.
- **Front Matter:** Title, project code, member names, student IDs, and institutional email addresses.
- **Visuals:** All tables and figures must be numbered, captioned, referenced within the body text, and cleanly rendered.
- **References:** Minimum of 25–30 academic references. Include DOI hyperlinks or permanent preprint URLs (e.g., arXiv) for all entries.
- **Data Consistency:** Every metric reported in the text must originate directly from reproducible log files within the repository.

5.2 Required Sections

#	Section	Detailed Description
1	Abstract	Problem statement, core agentic approach, quantitative highlights, and principal contributions (150–250 words).
2	Introduction	Contextual motivation, targeted research gaps, research questions, and explicit list of contributions.
3	Related Work	Synthesis of baseline literature, contextualizing the proposed architecture relative to existing work.
4	Problem Formulation	Formal mathematical or operational specification of the task, state space, tools, inputs, and outputs.
5	Proposed System	Complete architectural diagrams, agent roles, tool interfaces, memory structures, and gap components.
6	Experimental Setup	Benchmarks, data splits, baselines, LLM backend specifications, inference parameters, and tooling.
7	Evaluation Methodology	Definitions of quantitative metrics and formal procedure for comparative assessment.
8	Results	Empirical findings organized systematically by research gap, incorporating the gap-resolution table.
9	Ablation & Comparative Analysis	Component isolation studies, sensitivity analysis, token economics, latency, and runtime trade-offs.
10	Failure Analysis	Empirical distribution of failure types with in-depth trajectory walkthroughs of failures and edge cases.
11	Limitations & Validity	Methodological constraints, dataset boundaries, stochastic vulnerabilities, and generalization limits.
12	Conclusion	Summary of findings, architectural lessons, and recommended trajectories for future research.
13	References	25–30 accurately formatted academic citations with active DOI links.

6. Recommended Work Plan

Stage	Key Objectives & Milestones
Stage 1 — Build	Incorporate Phase 1 feedback; complete core agentic graph; implement gap modules; configure observability and data splits.
Stage 2 — Evaluate	Execute experiments E1–E6; perform ablation runs; log failure cases; refine system iteratively and update the dev log.
Stage 3 — Write & Submit	Compile paper draft from repository metrics; polish documentation and reproduction scripts; generate `phase2-final` release; complete submission.

7. Submission Guidelines

- **Deadline:** Sunday, 8 November 2026, 11:55 PM (PKT).
- **Submission Route:** Single upload per team by the group head via Google Classroom.
- **Compiled Paper:** Upload research paper PDF named: `Phase2_<ProjectCode>_<GroupHeadRollNo>_Paper.pdf` (e.g., `Phase2_G1-06_22I-1234_Paper.pdf`).
- **Paper Source:** Upload complete LaTeX source files as a ZIP archive.
- **Code Repository:** Include links to the public (or supervisor-shared) GitHub repository and the specific `phase2-final` release tag.

8. Assessment Rubric (100 Marks)

Evaluation is distributed across two core pillars: **Part A: Final Source Code, System Architecture & Reproducibility (40 Marks)** and **Part B: Research Paper, Evaluation & Academic Rigor (60 Marks)**.

#	Criterion	Marks	Excellent (90–100%)	Adequate (50–80%)	Weak (0–40%)
A1	Complete Agentic Pipeline	10	Runs end-to-end on full evaluation suite; genuine agentic reasoning, tool usage, and logged execution traces.	Functional pipeline with minor limitations in autonomy or trace logging.	Broken workflow or disconnected modules lacking agentic coordination.
A2	Gap-Specific Components	10	Dedicated, modular components addressing each target gap with clean interfaces for ablation.	Components present for some gaps or loosely integrated into the workflow.	Gaps unaddressed or components indistinct from standard baselines.
A3	Robustness & Iteration	5	Systematic error handling, retry/recovery loops, and clear dev log tracking problem-solution progression.	Basic error handling present; development history sparsely documented.	Brittle execution; no evidence of structured iterative improvement.
A4	Repository & Documentation	8	Professional architecture, clear installation/setup instructions, automated tests, no exposed secrets.	Working repository with incomplete documentation or disorganized layout.	Undocumented, non-functional, or inaccessible repository structure.

#	Criterion	Marks	Excellent (90–100%)	Adequate (50–80%)	Weak (0–40%)
A5	Reproducibility	7	One-click or fully scripted reproduction of all E1–E6 tables and metrics; tagged release created.	Most figures reproducible with manual intervention or missing script details.	Paper results cannot be reproduced from repository assets.
B1	Baseline & Base Comparison	8	Rigorous, fair comparison using identical data splits and metrics; reproduction status stated clearly.	Comparison present with minor split or configuration discrepancies.	No valid experimental baseline comparison included.
B2	Gap-Wise Experiments & Table	12	Comprehensive empirical tests for each gap hypothesis; thorough and transparent gap-resolution table.	Experiments conducted for all gaps but lacking analytical depth.	Superficial experiments; gap resolution omitted or unsupported.
B3	Ablation & Efficiency Analysis	7	Granular ablation isolating component contributions; detailed profiling of latency, costs, and token usage.	Partial ablation or cost analysis across single evaluation runs.	No ablation conducted; computational overhead unmeasured.
B4	Failure Analysis & Limitations	8	Categorized failure modes with quantified occurrences, agent trajectories, and thoughtful threat analysis.	Superficial failure review without trajectory inspection or categorization.	Failure analysis omitted; limitations unaddressed.
B5	Paper Structure & Content	12	All required sections present; sophisticated problem formulation, methodologies, and synthesis.	Most sections present; reads somewhat like a class project rather than a paper.	Key sections missing; insufficient technical depth throughout.
B6	Visuals & Results Presentation	7	Publication-grade figures and cleanly formatted tables matching logged data perfectly.	Readable tables and figures with minor aesthetic or formatting flaws.	Illegible, missing, or inconsistent tables and figures.
B7	Academic Writing & Referencing	6	Rigorous scholarly tone, proper IEEE/ACM format (8–10 pages), and 25–30 accurate references with DOIs.	Occasional formatting inconsistencies or minor bibliography gaps.	Informal tone, incorrect pagination, or insufficient references.

9. Submission Checklist

- Phase 1 supervisor feedback addressed and documented.
- End-to-end agentic workflow operational with comprehensive trajectory logging.
- Distinct, switchable modules implemented for each identified research gap.
- All experiments (E1–E6) executed on frozen data splits with statistical measures.
- Ablation analysis and failure mode trajectory investigations complete.
- Completed gap-resolution table included in both README and research paper.
- Clean repository with automated reproduction scripts and a `phase2-final` release tag.
- 8–10 page two-column IEEE or ACM paper with all 13 core sections and 25–30 DOI-referenced citations.
- LaTeX source (ZIP) and compiled PDF submitted by the group head prior to the deadline.